Advanced Industrial Organisation / Project 1

The European Car Market

Casper Moes (hwd362) 21. October 2026

Use of AI tools. I used ChatGPT (OpenAI) for code suggestions, numerical checks and first drafts of paragraphs based on my prompts. It also assisted with LaTeX formatting. I remain responsible for the final text, code and results, and for understanding the models used.

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1 Empirical part

1.1 Demand estimation

The data cover 40 cars in each of 150 country-year markets in Belgium, France, Germany, Italy and the UK, 1970-1999. I define market size as $M_t = \text{pop}_t/2.5$, approximating one annual purchase opportunity per household. Thus $s_{jt} = q_{jt}/M_t$ and $s_{0t} = 1 - \sum_j s_{jt}$. The outside option includes not purchasing a new car, buying used and buying an excluded model. Household size is an assumption; a larger M_t increases outside diversion and can reduce estimated merger harm. The changing top-40 sample also means excluded cars are imperfectly represented by one outside option.

1.1.1 Simple logit

In the simple logit model, utility of good j in market t for household i is given by

$$u_{ijt} = \underbrace{\mathbf{x}'_{jt}\boldsymbol{\beta} - \alpha \log p_{jt} + \xi_{jt}}_{\equiv \delta_{jt}} + \epsilon_{ijt}, \quad \epsilon_{ijt} \overset{iid}{\sim} T1EV(0). \tag{1.1}$$

Prices p_{ij} are `princ`: VAT-inclusive local prices divided by nominal GDP per capita. Controls are weight in tonnes, fuel consumption, $hp/100$ and a home-product dummy, together with brand and country fixed effects.

Country effects absorb persistent country demand differences; brand effects absorb permanent brand appeal. Price variation remains across models within brand-country partitions and as relative brand positions change over time. Thus unobserved model quality (i.e. ξ_{jt})), affecting brand appeal and time trends are the potential causes of endogeneity.

From (1.1) and by normalizing the outside options utility to zero (i.e. $\delta_{0t} = 0$) we can obtaion the choice probabilities

$$s_{jt} = \frac{\exp\{\delta_{jt}\}}{1 + \sum_{k \in \mathcal{J}_t} \exp\{\delta_{kt}\}}. \tag{1.2}$$

Observed shares are treated as these population probabilities (which they would be in the limit if (1.1) describes the utility well). This yields the regression model

$$\log s_{jt} - \log s_{0t} = \mathbf{x}'_{jt}\boldsymbol{\beta} - \alpha \log p_{jt} + \xi_{jt} \tag{1.3}$$

The model (1.3) is later estimated by both OLS and IV (2SLS), with standard errors clustered by car model code.

1.1.2 Nested logit

The simple logit estimation has two major problems. Firstly, the IIA assumption can be a poor representation of real-world substitution patterns and thus affect the merger simulation. One way to relax IIA is to introduce correlated match values within the five car classes: Subcompact, Compact, Intermediate, Standard and Luxury. The outside option is its own nest. Writing $g(j)$ for j's class, nested logit gives

$$D_{gt} = \sum_{k \in g} \exp\{\delta_{kt}/(1-\rho)\}, \qquad s_{jt} = \frac{\exp\{\delta_{jt}/(1-\rho)\} D_{g(j)t}^{-\rho}}{1 + \sum_g D_{gt}^{1-\rho}}, \quad 0 \le \rho < 1.$$

At $\rho = 0$ this reduces to simple logit. Nesting relaxes global IIA, while retaining proportional substitution conditional on class. Price sensitivity remains homogeneous. Demand inversion gives one linear regression:

$$\log s_{jt} - \log s_{0t} = \mathbf{x}'_{jt}\boldsymbol{\beta} - \alpha \log p_{jt} + \rho \log s_{j|g,t} + \xi_{jt}, \tag{1.4}$$

where $s_{j|g,t} = s_{jt} / \sum_{k \in g(j)} s_{kt}$.

1.1.3 Endogeneity

The second problem is price endogeneity: unobserved quality may raise both prices and demand. In nested logit, within-class share also responds to unobserved quality. I therefore instrument log price in simple logit and *both* log price and log within-class share in nested logit.

Both IV models use the same two instruments: the number of other cars in the same class made by the same manufacturer, and the number made by rival manufacturers. Counts are calculated within each country-year market and exclude the car itself. They describe the competitive environment, which helps predict prices and within-class shares.

The instruments are valid only if these counts are unrelated to unobserved demand. This may fail if manufacturers choose their portfolios in response to expected demand or unobserved quality. Country and brand effects absorb persistent differences, but not changing quality or common demand shocks. Observed characteristics are still treated as exogenous, so IV does not necessarily solve every endogeneity problem.

1.2 Estimation results

The above explained estimations lead to the results displayed in 1 which we will discuss now. In simple logit, α increases from 1.145 under OLS to 2.393 under IV and thus

the price sensitivity increases. [1] In nested logit, it increases from 1.163 to 4.114. This is consistent with unobserved quality raising both prices and demand, causing OLS to understate the negative price effect. However, this interpretation depends on the instruments being valid.

Nested OLS estimates $\rho = 0.658$, implying substantial substitution within classes. Nested IV reduces this estimate to 0.168, with a standard error of 0.140. Thus, instrumenting also changes the estimated substitution pattern, but the IV evidence for a departure from simple logit is limited.

Home-product coefficients remain positive in all four models, ranging from 0.544 to 1.253. This suggests sizeable home bias after controlling for observed characteristics and fixed effects. It need not reflect a pure causal preference: domestic distribution networks or unobserved country-specific quality may still matter.

1.3 Merger analysis

A given firm / car manufacturer f in market t, with product portfolio $\mathcal{J}_{ft}$, maximises $\sum_{k \in \mathcal{J}_{ft}} M_t(p_{kt} - c_{kt}) s_{kt}(\mathbf{p}_t)$. Here $p = p^C/(1 + tax)$ is producer price; taxes are held fixed. Stacking the FOCs for all firms, with $[\nabla \mathbf{s}]_{jk} = \partial s_k / \partial p_j$, gives

$$\mathbf{s}_t(\mathbf{p}_t^*) = -[\mathcal{H}_t \odot \nabla \mathbf{s}(\mathbf{p}_t^*)](\mathbf{p}_t^* - \mathbf{c}_t), \qquad \widehat{\mathbf{c}}_t = \mathbf{p}_t^0 + [\mathcal{H}_t^0 \odot \nabla \mathbf{s}(\mathbf{p}_t^0)]^{-1} \mathbf{s}_t^0. \tag{1.5}$$

The ownership matrix has $\mathcal{H}_{jk} = 1$ if j and k have the same owner, and zero otherwise. Firms compete on price and all optimise given the other firms' prices. These inferred costs depend on static multiproduct Bertrand conduct and constant marginal costs.

To simulate VW-Fiat, I link their products in $\mathcal{H}_t^1$, retain these costs and let all firms reoptimise. I use PyBLP's Morrow-Skerlos iteration for every market and all four models. All 600 baseline cases have positive costs and pass independent pricing and profit checks.

The diversion ratio is the fraction of sales lost by j following a small own-price increase that moves to k:

$$D_{j \to k,t} = -\frac{\partial s_{kt} / \partial p_{jt}}{\partial s_{jt} / \partial p_{jt}}, \qquad E_{jk,t} = \frac{\partial s_{jt}}{\partial p_{kt}} \frac{p_{kt}}{s_{jt}}.$$

For simple logit, $D_{j \to k,t} = s_{kt}/(1 - s_{jt})$, so OLS and IV have identical diversion at observed shares. Nested logit allows stronger within-class substitution. I sum diversion from each VW car to Fiat cars, and conversely, weighting source cars by premerger

[1] Cluster-robust Wald tests reject joint insignificance of the two excluded instruments: $\chi^2(2) = 24.35$ ($p = 5.15 \times 10^{-6}$) for log price and $\chi^2(2) = 272.78$ ($p = 5.84 \times 10^{-60}$) for log within-class share. Table 1 reports Wald/2, not conventional F-statistics. Nested IV is exactly identified here, so no overidentification test is available.

sales within market, then averaging equally over markets. Table 2 reports median own elasticities and mean cross elasticities between distinct cars in the same class.

1.3.1 Effects of the merger

Table 2 reports the average price increase for the combined VW-Fiat portfolio. Instrumenting reduces this increase from 4.31% to 0.45% in simple logit and from 3.31% to 0.51% in nested logit. The IV estimates imply greater price sensitivity: median own-price elasticities are -2.39 in simple IV and -4.88 in nested IV.

Class nesting increases substitution between VW and Fiat: partner diversion rises from 0.56% in simple IV to 1.77% in nested IV. However, nested IV also implies more elastic own demand and lower inferred price–cost margins, which fall from 42.5% to 21.6% for the merging firms. These changes offset much of the stronger incentive to recapture sales, so the predicted price increase rises only from 0.45% to 0.51%. Under OLS, partner diversion increases from 0.56% to 5.66%, but inferred margins fall from 88.9% to 37.7%, and the price increase falls from 4.31% to 3.31%. Greater diversion therefore does not determine the merger effect alone: demand sensitivity, margins and demand curvature also matter.

Figure 1 shows that Italy has the largest price increase and the UK the smallest under all four models, with Germany, Belgium and France between them. Under nested IV, the increase is 0.91% in Italy and 0.24% in the UK. Partner diversion is also higher in Italy: 2.97%, compared with 0.91% in the UK. This indicates greater competitive overlap between VW and Fiat in Italy, giving the merged firm a stronger incentive to raise prices.

Consumer welfare is measured using the logsum $L_t^h = \log[1 + \sum_g (D_{gt}^h)^{1-\rho}]$, where $h = 0$ denotes the premerger equilibrium and $h = 1$ the postmerger equilibrium. With log prices, dividing the utility change by α does not yield monetary consumer surplus. I instead report the uniform percentage increase in all baseline car prices that would generate the same utility loss:

$$W_t = 100 \left[\left(\frac{A_t^0}{A_t^1} \right)^{1/\alpha} - 1 \right], \qquad A_t^h = e^{L_t^h} - 1, \qquad \overline{W} = \frac{1}{T} \sum_t W_t.$$

Positive values indicate consumer harm. The average welfare loss falls from 1.52% to 0.17% when instrumenting simple logit, and from 1.29% to 0.19% when instrumenting nested logit. These figures are price equivalents, not percentage changes in consumer surplus.

To assess potential efficiencies, I reduce VW-Fiat marginal costs to $(1 - \kappa)c_{jt}$, leave rivals' costs unchanged and solve the pricing equilibrium again. For each market, I find the saving that restores baseline expected utility. Taking the largest of these thresholds

gives one common saving sufficient to offset consumer harm in every market. The required reductions are 1.96% under simple IV and 1.90% under nested IV, compared with 27.87% and 21.73% under their OLS counterparts. Strict welfare gains require savings above these thresholds. This criterion protects expected welfare in each market, not every household. Without verified efficiencies, the estimates do not support approval on consumer-welfare grounds; the variation across models also cautions against treating the thresholds as precise policy cutoffs.

The robustness checks reveal some sensitivity. Defining market size as population/2 instead of population/2.5 reduces the nested-IV price effect from 0.51% to 0.47%, while all four models still produce prices in all 150 markets (Table 3). Population/3 produces one negative simple-OLS marginal cost, so that model's merger average is not reported. Adding year effects also invalidates simple-OLS cost recovery, showing that the baseline comparison is sensitive to omitted time effects. Adding a third count instrument gives similar IV price predictions, but does not establish instrument validity. Endogeneity, instrument selection and the assumptions of static price competition and constant marginal costs remain limitations.

2 Theoretical part

2.1 Price dispersion for homogeneous goods

The classic price-dispersion models considered can all be slightly augmented to explain the qualitative pattern of home bias. For example, search models with homogeneous goods, where we exogenously set the information cost to favour home goods in some way.

Take for example Diamond (1971), where a unit continuum of consumers value the homogeneous good at $v > c$, where c is the firms' constant marginal cost. Consumers first search at a randomly selected firm at zero cost. Then they choose whether or not to search another random firm, incurring a search cost $s > 0$. The unique symmetric pure-strategy Nash equilibrium (PSNE) is that firms choose monopoly prices $p^* = v$ and consumers only search the first firm assigned to them. Intuitively, all other potential equilibria with $p^* < v$ incentivise a firm deviation of

$$p' = \min\{p^* + s - \varepsilon, v\}, \qquad 0 < \varepsilon < s.$$

To see this, let a consumer i search the deviating firm with price p' at zero cost. They can either buy this good at cost $p' \leq p^* + s - \varepsilon$ or search and buy a non-deviating firm's good at total cost $p^* + s$. Since goods are homogeneous they strictly prefer buying at p' to searching and buying at p^*.

Consider Diamond (1971) slightly augmented so the initial zero-cost search is drawn

from a distribution that assigns greater probability to domestic firms while remaining uniform within the domestic and foreign groups. If no domestic firms are available, the assignment is uniform across all firms. This would induce an equilibrium with home bias as observed in the data.

Much the same analysis could be conducted for many other models by biasing the exogenous parameters towards home goods (i.e. uninformed customers in Varian (1980), search costs in Burdett and Judd (1983) or advertising costs Butters (1977)).

2.2 Perloff-Salop and logit

In the Perloff-Salop model consumers have utility given by

$$u_{ij} = \epsilon_{ij} - p_j, \tag{2.1}$$

where $\epsilon_{ij} \sim F$ are independent and identically distributed stochastic match values, interpreted as how well product j fits consumer i's preferences.

The distribution F is arbitrary and-crucially for this discussion-does not depend on observable characteristics of j. Therefore, product differentiation comes from the random match values, rather than observed characteristics.

Looking back at the simple logit, we see that utility does depend on j's characteristics through $\mathbf{x}_{jt}'\boldsymbol{\beta}$. This is the main difference; the empirical models also use log prices and a finite outside option. In (1.1) the match-value distribution is chosen as the Type I extreme value distribution (with location zero). Nested logit additionally correlates match values within each class, while retaining common price sensitivity.

2.3 Testing the Perloff-Salop model

The distribution F in the Perloff-Salop model is by no means irrelevant (as I will show in Section 2.4), and therefore whenever empirical models of the form (2.1) are tested, the distributional choice of T1EV(0) is part of the joint hypothesis.

In addition to this, as always, exogeneity of the regressors must hold; the regression test cannot establish this assumption.

Having this in mind, we test the following null hypothesis:

$$\mathcal{H}_0 : \boldsymbol{\beta} = \mathbf{0},\ \gamma = \mathbf{0}, \qquad \mathcal{H}_A : \text{at least one characteristic or brand effect is nonzero.}$$

Here the null includes the four characteristics and brand effects; for this linear test, market effects are retained, allowing common market demand shifts and a finite outside option. Rejecting it would imply that observable characteristics affect utility, contradicting this adjusted version of (2.1). Estimating (1.2) and using a cluster-robust F-test

yields $F(39, 284) = 53.005$ and $p < 0.001$. Thus we firmly reject the joint null, rather than the entire family of models with arbitrary F.

2.4 Many car manufacturers

Using the result by Gabaix, Laibson and Resnick (2016), markups in the Perloff-Salop model have the asymptotic property, as the number of firms $n \to \infty$,

$$p^* - c \propto \frac{1}{nf[F^{-1}(1 - 1/n)]}. \tag{2.2}$$

Here f is the density of the i.i.d. taste shocks and F their cdf. Inserting the two distributions respectively and letting $n \to \infty$ informs us about the markup with many car manufacturers.

2.4.1 T1EV shocks

Here I restate the distribution for clarity:

$$F(x) = \exp\{-\exp\{-x\}\}, \qquad f(x) = \exp\{-x\}F(x).$$

Inserting into the denominator of (2.2) yields

$$\begin{aligned} nf[F^{-1}(1 - 1/n)] &= n \exp\{-F^{-1}(1 - 1/n)\}(1 - 1/n) \\ &= -n \log(1 - 1/n)(1 - 1/n) \longrightarrow 1. \end{aligned}$$

Taking the limit uses $\lim_{n\to\infty}(1 - 1/n)^n = \exp\{-1\}$. That is, when taste shocks follow a T1EV(0) distribution, increasing the number of firms indefinitely does not remove the markups.

2.4.2 Gaussian shocks

For Gaussian shocks, let $F = \Phi$ and $f = \phi$. Set $x = \Phi^{-1}(1 - 1/n)$, so $n = 1/[1 - \Phi(x)]$. As $n \to \infty$, we have $x \to \infty$.

Using l'Hôpital's rule and $\phi'(x) = -x\phi(x)$,

$$\lim_{n\to\infty} n\phi[\Phi^{-1}(1 - 1/n)] = \lim_{x\to\infty} \frac{\phi(x)}{1 - \Phi(x)} = \lim_{x\to\infty} \frac{-x\phi(x)}{-\phi(x)} = \infty.$$

The denominator in (2.2) therefore grows without bound. Its reciprocal approaches zero, so markups vanish and prices approach marginal cost, unlike the T1EV case.

The calculation above uses the lecture's level-price utility. In the empirical log-price model, symmetric Gumbel shares $1/n$ instead give $(p - c)/p = 1/[\alpha(1 - 1/n)]$, so

$p/c \to \alpha/(\alpha - 1)$ for $\alpha > 1$. Gaussian shocks give a vanishing Lerner index. Thus the contrast survives, but the price transformation changes the markup's units.

2.4.3 Implications for empirical estimates

The above results highlight that the assumed taste-shock distribution is by no means innocent in a Perloff-Salop model. This hints at a more general insight. The assumptions made to derive logit (for example the T1EV assumption on taste shocks) are not innocent and can themselves cause certain regularities of the models. These regularities or outcomes are not findings but assumptions and should be stated as such.

Another example could be the choice of nests in a nested logit model: putting cars in the same class makes them closer substitutes by assumption. This does not mechanically increase estimated merger harm, because price sensitivity and recovered margins also change. Here we again see that the modelling decisions before estimation partly determine the results.

3 Sources

Supplied Project 1; Munk-Nielsen, *Demand Estimation* (2026), Sections 1-5; Vikander, Lectures 1, 5-6 (2026); Goldberg-Verboven (2001) data.

4 Results / Output

Table 1: Demand estimates: log prices and nests on class

	Simple OLS	Simple IV	Nested OLS	Nested IV
Price α	1.145 (0.139)	2.393 (0.628)	1.163 (0.104)	4.114 (1.553)
Nesting ρ	-	-	0.658 (0.026)	0.168 (0.140)
Weight	1.300 (0.254)	1.863 (0.402)	0.175 (0.162)	2.351 (0.629)
Fuel consumption	−0.101 (0.019)	−0.031 (0.039)	−0.072 (0.013)	0.072 (0.090)
Power	−0.322 (0.250)	0.492 (0.498)	−0.016 (0.185)	1.689 (1.102)
Home product	1.238 (0.050)	1.253 (0.048)	0.544 (0.063)	1.096 (0.148)
First stage: price	-	12.2	-	12.2
First stage: within share	-	-	-	136.4

Note: 6,000 observations, 150 markets. Model-clustered SEs in parentheses (285 clusters), without finite-sample correction. All four models include additive brand and country effects, weight/1000, power/100, fuel and home. Only price is logged. Both IV models use the same two count instruments. First-stage entries are clustered Wald statistics divided by two; they are marginal diagnostics, not conditional identification tests. Nested IV instruments both price and within-class share.

Table 2: VW-Fiat merger: all four models cover all 150 markets

	Simple OLS	Simple IV	Nested OLS	Nested IV
Combined price increase (%)	4.31	0.45	3.31	0.51
Median own elasticity	−1.14	−2.39	−3.23	−4.88
Mean cross: within class	0.0022	0.0046	0.2083	0.0844
Partner diversion (%)	0.56	0.56	5.66	1.77
Welfare price equivalent (%)	1.52	0.17	1.29	0.19
Required cost saving (%)	27.87	1.96	21.73	1.90

Note: Prices and partner diversion use premerger unit-sales weights within markets, then equal market weights. Welfare is a uniform-price equivalent, not money CS or a percentage of CS. Savings reduce only VW-Fiat marginal costs and restore expected welfare in every market; strict gains require exceeding the threshold. Full precision is retained in the code.

Table 3: Market-size check: population/2 versus population/2.5

	Combined price increase (%)	
Model	Baseline	Pop./2
Simple OLS	4.31	3.56
Simple IV	0.45	0.36
Nested OLS	3.31	3.30
Nested IV	0.51	0.47

Note: All models are re-estimated; shares, inferred costs and mergers are recomputed in all 150 markets. The top panel reports the merging portfolio's sales-weighted price change; the bottom panel reports lost sales diverted to the partner. Markets receive equal weights.

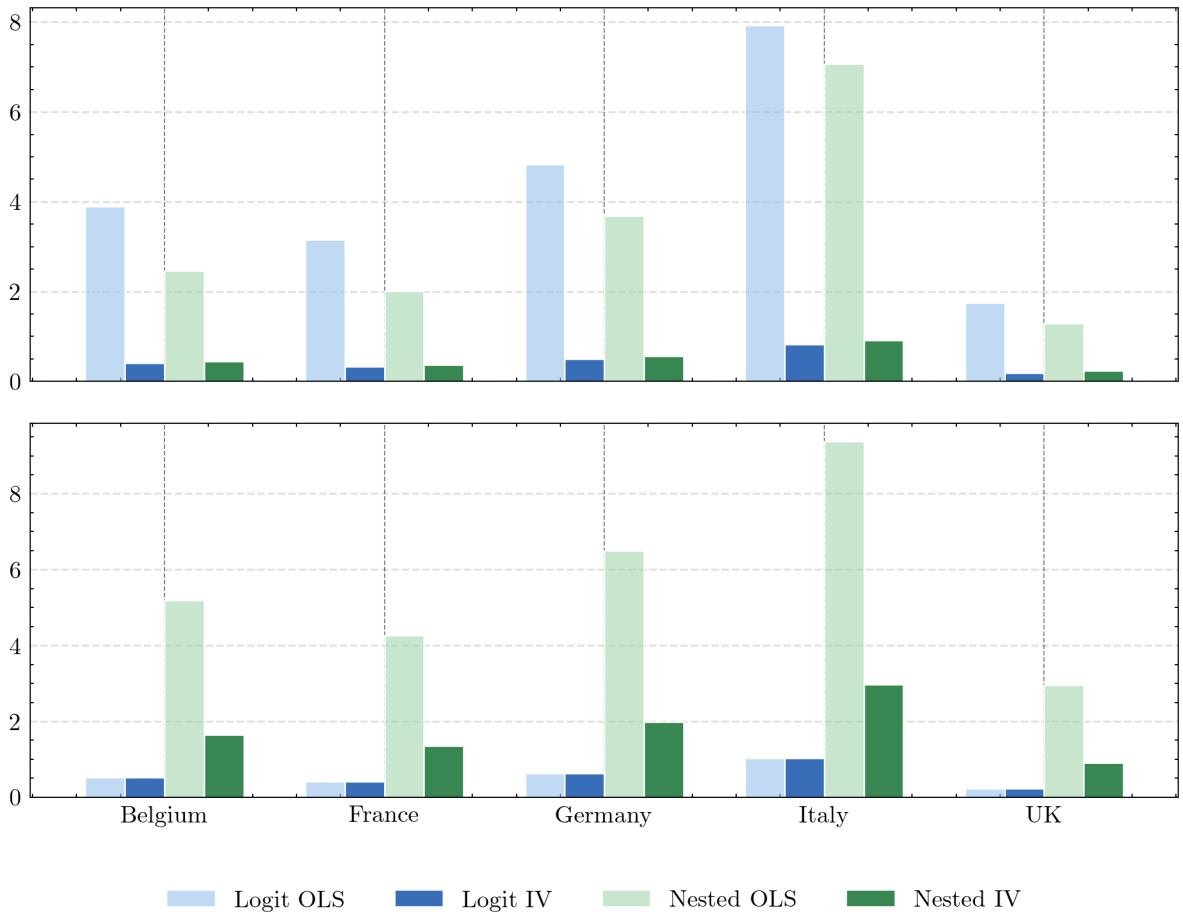


Figure 1: Merger price effects and partner diversion by country